

Signals and Systems — Tutorial 3

Fourier Series

Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a ★.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

Pework questions

A: Frequency and Spectra

Suppose $f(t) = \sin(t) + \cos(3t)$ find the fundamental frequency of f and plot its magnitude and phase spectrum.

B: Fourier Series

Determine the complex Fourier series for 1. $f(t) = \sin(\omega t) + \cos(\omega t)$ and 2. $f(t) = \sin(\omega t) \cos(\omega t)$

C: MATLAB

Suppose $f(t)$ has a Fourier expansion

$$f(t) = 2 \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} \sin((k+1)t)$$

1. Without calculation or plotting is $f(t)$ even, odd or neither? Explain.
2. Using MATLAB (or your favourite programming language) plot $f(t)$ for $k = 0, 1, \dots, 5$. What function is $f(t)$? *You do not have to include a copy of the plot in your prework submission.*

Tutorial questions

1: Fourier Series

Find the Fourier series of **(a)** each of the periodic signals in Figure 1 (**Hint:** Use properties of the Fourier series) and **(b)** of the rectified sine-wave $f(t) = |\sin(100t)|$ which was the output from the ‘full-wave rectifier’ in Question 5 in Tutorial 1.

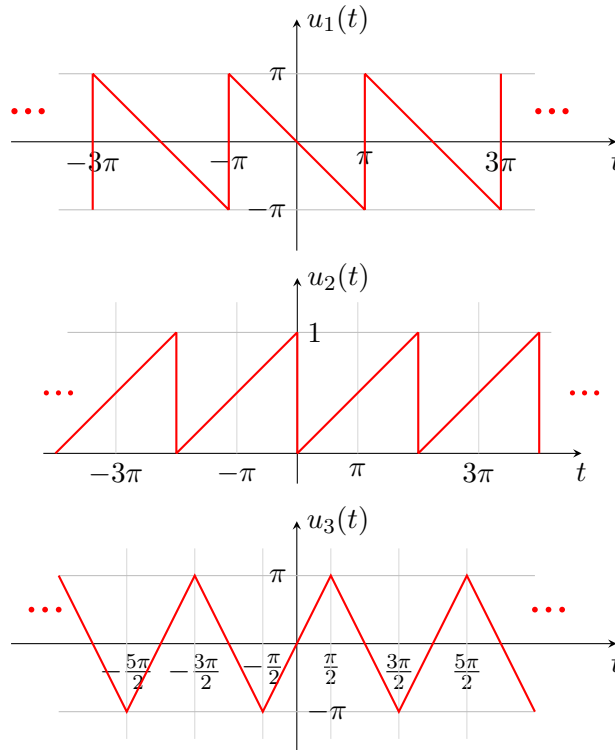


Figure 1: Periodic Signals

2: Exponential and Trigonometric Series

- From the definitions given in the lecture notes, find both the exponential and trigonometric Fourier series for at least one of the periodic signals given in Figure 1.
- Use Euler's formula to prove that

(a) $c_0 = a_0/2$,

(b) $|c_k| = \sqrt{a_k^2 + b_k^2}$,

(c) $2c_k = a_k + ib_k$ and

(d) $\bar{c}_k = c_{-k}$, where $\bar{c}_k$ is the complex conjugate of the coefficient c_k .

and verify the above identities for the coefficients a_k, b_k and c_k of the series found in Part 1.

3. Recall from Lecture 4 that the Fourier series for the square-wave shown in Figure 2 is

$$u(t) = \sum_{k \text{ odd}} 2 \operatorname{sinc}(k/2) \cos(kt)$$

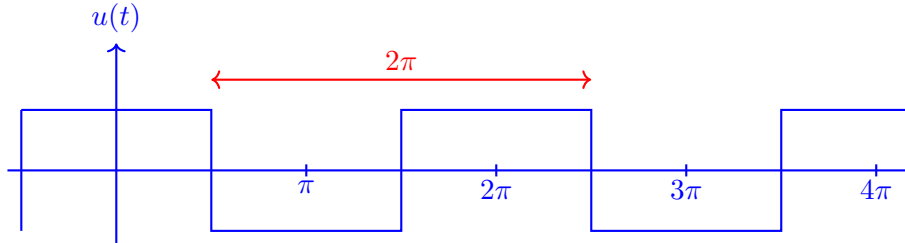


Figure 2: Square-wave with period $T = 2\pi$ and amplitude 1

Consider the periodic square wave $f(t)$ with period $T = 2\pi$ and amplitude ± 1 . Compute the Fourier series coefficients c_k for $k = 0, 1, 2, 3$

3: Spectra

1. Plot the magnitude and phase spectrum for at least one of the signals in Figure 1.
2. For the spectrum in Figure 3 find the Fourier expansion and determine if the function is odd, even or neither **Hint:** Consider the spectra of $\sin(\omega t)$ and $\cos(\omega t)$.

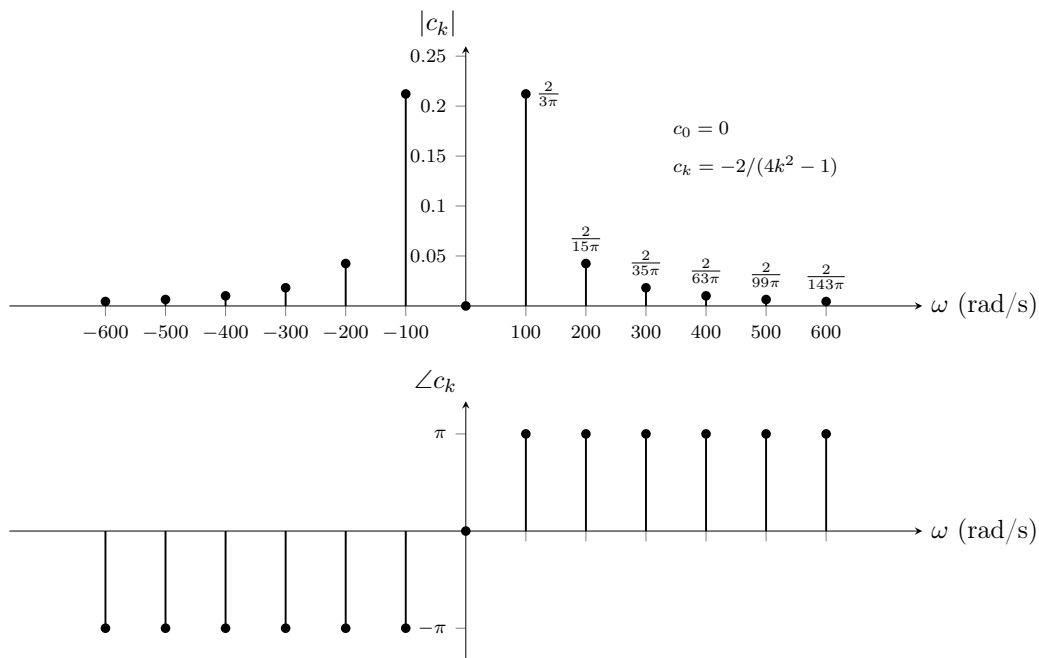


Figure 3: Magnitude and phase spectrum for an unknown signal.

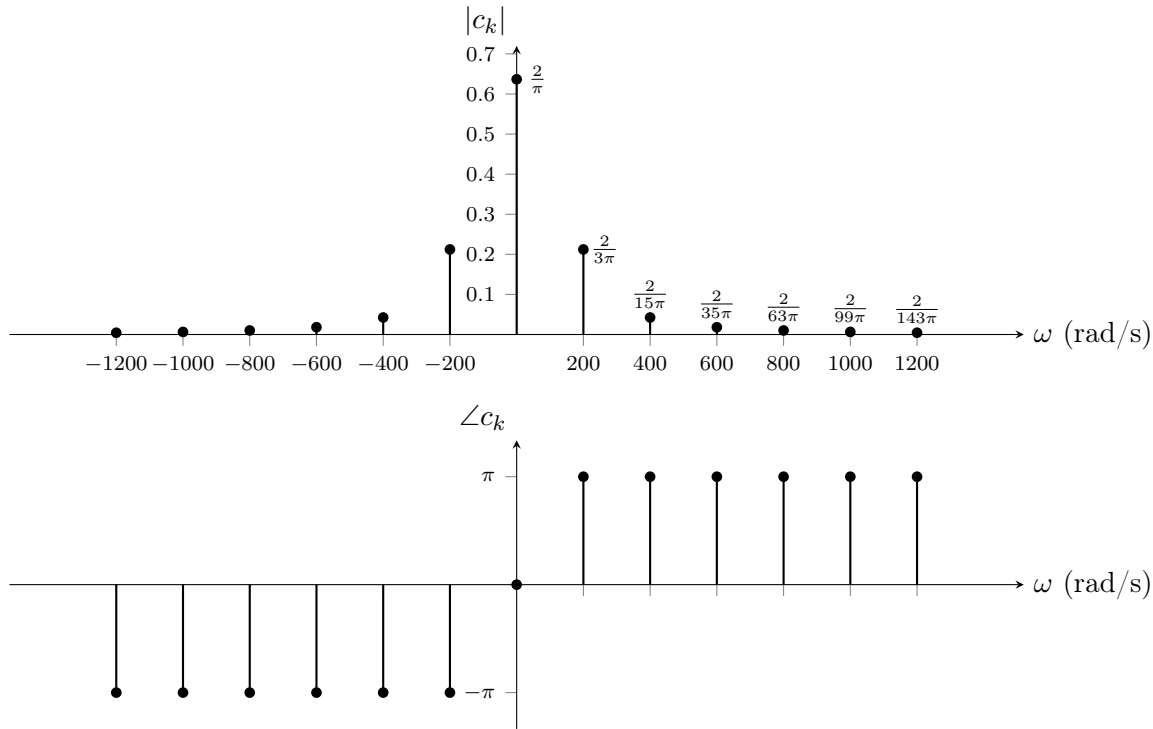


Figure 4: Magnitude and phase spectrum for $|\sin(100t)|$.

3. Given the magnitude and phase spectrum of $f(t) = |\sin(100t)|$ is as shown in Figure 4 determine which function is shown in Figure 3.

4: Parseval's Theorem

1. Given that the area of a polar function $r(\theta)$ is

$$A = \frac{1}{2} \int_a^b r^2(\theta) d\theta$$

Find the area of (a) $f(t) = \cos(t) + \sin(t)$ and (b) $f(t) = 1 - \cos(t)$

- ★ 2. **[Extension]** Using the Fourier expansion of $f(t) = t$ (see the prework questions). Prove that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

5: Differentiation and Fourier Series

1. **[Extension]** Consider the circuit in Figure 5

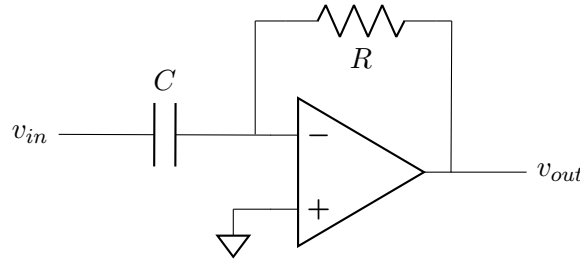


Figure 5: Ideal differentiator

Verify that the input $u(t) = v_{in}(t)$ and output $y(t) = v_{out}(t)$ satisfy

$$y(t) = -RC \frac{d}{dt} u(t)$$

2. Assume $RC = 1$. Suppose $u(t) = u_3(t)$ is the triangular wave given in Figure 1. Verify that $y(t)$ is a square-wave with amplitude 1 and the same frequency as the square-wave given in Figure 2.
3. Consider the Fourier series for a triangular wave $u_3(t)$ and square-wave in Figure 2 i.e.

$$u_3(t) = \sum_k c_k e^{jk\omega t} \text{ and } u_{sq}(t) = \sum_k d_k e^{jk\omega t}$$

- (a) By differentiating term-wise, verify that $u_3(t) = u_{sq}(t)$
- (b) **[Extension]** Find d_k/c_k for each k . How does this compare with the frequency response function (equivalent impedance), $Z_{eq} = R/(j\omega C)$, of the circuit in Figure 5?
- ★ 4. Find the Fourier series for the signal $u_\delta(t) = \delta(t - n\pi)$ shown in Figure 6 where $n \in \mathbb{Z}$.

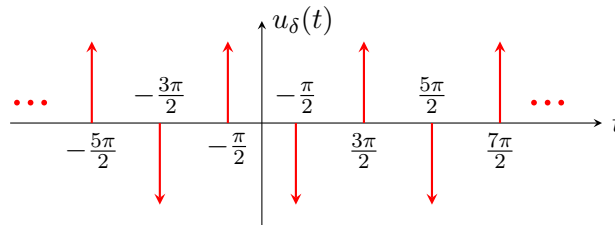


Figure 6: $\delta(t - n\pi)$